

Extended Results and Diagnostics for the Paper Moving Beyond Crude Trial–Real-World Comparisons Using Counterfactual Marginal Alignment

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This document accompanies the paper *Moving Beyond Crude Trial–Real-World Comparisons Using Counterfactual Marginal Alignment* and provides expanded methodological details, additional analyses, and more comprehensive discussion of the results than is feasible in the main article.

1 Results

This section presents the results of the real-world and trial comparisons in stages corresponding to the analytical framework described above. We first summarize results from the multilevel outcome models fitted to the real-world cohort, which characterize associations between baseline variables and pCR. We then compare observed pCR rates between the real-world cohort and the randomized trial arms and present counterfactual estimates obtained using MAIC and Bayesian MRP with raking. Finally, we decompose the baseline-adjusted component of the efficacy–effectiveness gap into variable-specific contributions.

All results are descriptive and address transportability rather than causal treatment effects. Adjustment is restricted to baseline variables that are observed and harmonizable across the real-world and trial populations and therefore does not ensure unconfounded comparisons, either within individual RCT arms or across trials. Limited covariate overlap and unmeasured confounding remain plausible and are captured in the residual component of the analysis.

1.1 Results of the Outcome Models

Each real-world–trial comparison uses a distinct set of baseline variables, resulting in six multilevel logistic regression models: three models based solely on baseline characteristics and three that additionally include an indicator of

Table 1: Baseline characteristics of the HUS cohort.

Characteristic	N	Summary
Median age (n = 268)	—	52 (42–63)
Tumor size, median (mm)	221	26.0 (20–37)
Tumor size ≤ 20 mm	58	22% [17–27%]
Tumor size 20–50 mm	143	53% [47–60%]
Tumor size > 50 mm	19	7% [4–11%]
Tumor size: unknown	47	18%
Nodal status: positive	155	58% [52–64%]
Nodal status: negative	93	35% [29–41%]
Nodal status: unknown	20	8% [5–11%]
Bilateral breast cancer	8	3% [1–6%]
Unilateral breast cancer	260	97% [94–99%]
Bilaterality: unknown	0	—
Histology: ductal	211	79% [73–84%]
Histology: lobular	13	5% [3–8%]
Histology: other	44	16% [12–21%]
Histology: unknown	0	—
Tumor grade 1	< 3	$< 1\%$ [0–2%]
Tumor grade 2	64	24% [19–29%]
Tumor grade 3	152	57% [51–63%]
Tumor grade: unknown	51	19% [15–24%]
ER and/or PR: positive	88	33% [27–39%]
ER and/or PR: negative	180	67% [61–73%]
ER and/or PR: unknown	0	—
HER2 IHC/ISH: IHC 2+ and ISH+	71	27% [21–32%]
HER2 IHC/ISH: IHC 3+	183	68% [62–74%]
HER2 IHC/ISH: other	14	5% [3–9%]
HER2 IHC/ISH: unknown	0	—
Ki-67, median (n = 263)	—	40 (40–60)
Ki-67 $\leq 30\%$	91	34% [28–40%]
Ki-67 $> 30\%$	172	64% [58–70%]
Ki-67: unknown	5	2% [0–4%]

Notes: Continuous variables are reported as median (interquartile range). Categorical variables are reported as n (%) with 95% binomial confidence intervals. Counts between 1 and 2 are masked as “ < 3 ” to preserve patient confidentiality.

Abbreviations: ER, estrogen receptor; PR, progesterone receptor; HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; ISH, in situ hybridization; Ki-67, proliferation index.

treatment discontinuation. This structure allows assessment of the robustness of estimated associations to model specification.

These outcome models are not intended to estimate causal effects or to provide optimal predictive performance. Instead, covariate selection and discretization are guided by the objective of harmonizing marginal distributions between the real-world and trial populations. Posterior summaries of coefficients for all six models are reported in Supplementary Tables 5–11. All the coefficients are identified via a sum-to-zero constraint and should be interpreted as deviations from the overall mean of the linear predictor. Pairwise differences between categories are obtained by subtracting the corresponding coefficients.

Despite their descriptive role, several baseline variables show consistent associations with pCR across model specifications. Pre-treatment hormone receptor status is among the most strongly associated variables. As an example in the HELEN-006 outcome model (Supplementary Table 9), hormone receptor positivity has a mean posterior coefficient of -0.64 with a 94% highest density interval (HDI) from -0.93 to -0.35 , which is consistent over all the models. Using the “divide-by-4” rule as a rough approximation, this coefficient of -0.64 corresponds to an upper bound of approximately 16 percentage points difference in pCR rates between hormone-positive and hormone-negative patients, holding other variables constant. The magnitude of this association is broadly consistent with published results in similar patient populations, while lying near the upper end of reported estimates [1, 7, 6, 2, 4].

Besides the negative hormone receptor status, higher HER2 expression is also associated with higher modeled pCR rates in the HUS sample. For example, comparing IHC3+ to IHC2+ corresponds to an approximate difference of 35 percentage points under the same approximation. Some other coefficients show less intuitive patterns, such as T3 tumors having comparable or higher modeled pCR rates than T2 tumors, which may reflect residual confounding or missing interactions between the variables.

Individual coefficients should be interpreted cautiously. Some variables in the HELEN-006 model are structurally related (e.g., stage, tumor size, and nodal status), which may introduce collinearity and reduce interpretability of individual parameters. This issue affects interpretation of individual coefficients but do not materially influence the post-stratified predictions that form the basis of the transportability analysis. In addition, many posterior intervals include zero, reflecting substantial uncertainty about the sign of the coefficient, particularly for imbalanced variables such as bilaterality.

Coefficients for NAT discontinuation are modest in magnitude in KRISTINE and HELEN-006 variable outcome models (Supplementary Table 8 and Supplementary Table 10), suggesting limited association with modeled pCR rates. However, in the NeoSphere variable outcome model, discontinuation is defined using a lower treatment threshold than in the KRISTINE and HELEN-006 comparisons, which may partly explain the larger magnitude (more negative) but more uncertain coefficient estimates observed in that setting.

1.2 Comparison of pCR Rates Across RCT, Real-World, and Counterfactual Populations

We next compare pCR rates between the HUS cohort and each RCT arm, considering both observed and baseline-adjusted estimates. For each trial, we present (i) the unadjusted comparison, (ii) the counterfactual estimate obtained by aligning baseline distributions, (iii) the difference between RCT and real-world pCR rates based on these estimates, and (iv) Shapley decomposition of the adjustment difference $\Delta\hat{\theta}_{\text{adj}}$. The same analytical framework is applied separately to the NeoSphere, KRISTINE, and HELEN-006 comparisons.

1.2.1 NeoSphere comparison

We first estimate unadjusted posterior distributions for pCR rates in the HUS cohort and in each RCT arm using binomial models. Posteriors are obtained from the observed sample sizes and numbers of patients achieving pCR in each group. A uniform Beta(1, 1) prior is used, as the likelihood dominates given the sample sizes.

For the HUS cohort, the posterior mean pCR rate $\hat{\theta}_{\text{RWD}}$ is 0.54 with a 94% HDI from 0.49 to 0.60. For the NeoSphere arm B group, the posterior mean is 0.46 with a 94% HDI from 0.37 to 0.55. The posterior for the observed difference $\Delta\hat{\theta}_{\text{obs}} = \hat{\theta}_{\text{RCT}} - \hat{\theta}_{\text{RWD}}$ has mean -0.09 and a 94% HDI from -0.19 to 0.02 , indicating that without baseline adjustment, the posterior mass favors credibly higher pCR rates in the HUS cohort (with 94% of the posterior mass).

We then adjust for differences in baseline marginals using Bayesian MRP with raking and MAIC. Under MRP with raking, the counterfactual HUS pCR rate has posterior mean 0.51 with a 95% HDI from 0.42 to 0.59. The corresponding difference $\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}} - \hat{\theta}_{\text{RCT}}$ is positive in 79% of posterior draws. The estimated posterior distributions of the pCR rates for HUS ($\hat{\theta}_{\text{RWD}}$), NeoSphere ($\hat{\theta}_{\text{RCT}}$), and the counterfactual ($\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}}$) are shown in Figure 2.

The MAIC-adjusted pCR estimate $\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}}$ is 0.58 with a bootstrapped 95% confidence interval of (0.46, 0.67). The implied difference relative to the RCT estimate is directionally consistent with the MRP with raking results. However, the MAIC-adjusted pCR rate is higher than both the MRP-based estimate and the unadjusted HUS pCR rate $\hat{\theta}_{\text{RWD}}$. This upward shift should be interpreted cautiously, as the effective sample size (ESS) under MAIC is low (bootstrapped mean ESS ≈ 30), which means that individual rows have higher weights and lead to increased variability in the estimates: see the weight distribution diagnostics in Supplementary Figure 1.

The posterior distributions of the adjustment difference $\Delta\hat{\theta}_{\text{adj}}$ and the residual difference $\Delta\hat{\theta}_{\text{res}}$ are shown in Figure 1. The posterior mean of the adjustment difference is -0.03 , with a 95% HDI from -0.13 to 0.08 . The posterior probability that the adjustment effect is negative is approximately 75%, while the probability that its magnitude exceeds 5 percentage points is 39%.

Among the three comparisons, the NeoSphere analysis is the only one in which the residual difference $\Delta\hat{\theta}_{\text{res}}$ is predominantly negative, with 79% of pos-

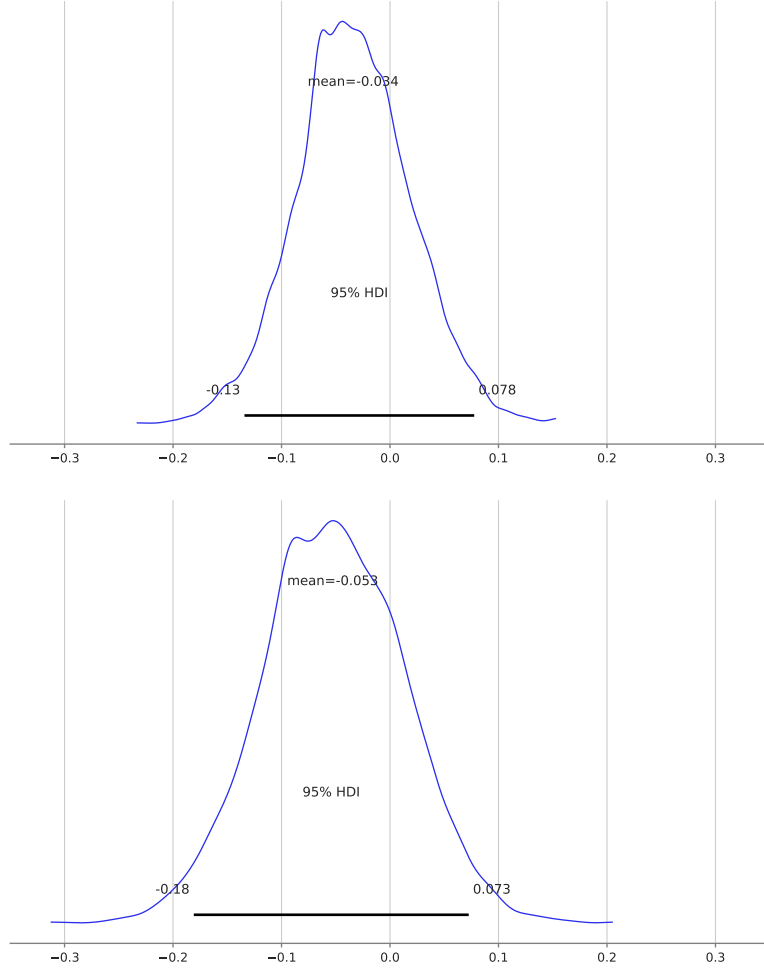


Figure 1: Posterior distributions of the adjustment difference $\Delta\hat{\theta}_{\text{adj}}$ (top) and residual difference $\Delta\hat{\theta}_{\text{res}}$ (bottom) for the NeoSphere comparison.

terior mass below zero. The posterior for $\Delta\hat{\theta}_{\text{res}}$ is displayed in the lower panel of Figure 1. By construction, this residual component aggregates differences not accounted for by the baseline adjustment, including treatment differences and unmeasured baseline factors. In the NeoSphere case, plausible contributors include differences in disease staging (see Supplementary Table 2) and the shorter NeoSphere protocol of four THP cycles compared with the six cycles typically used in HUS. The magnitude of the residual difference is modest, which should be taken into account when interpreting this posterior distribution: the posterior probability that the counterfactual HUS pCR rate exceeds the NeoSphere rate by more than 5 percentage points is 52%.

The Shapley decomposition of $\Delta\hat{\theta}_{\text{adj}}$ reflects the overall uncertainty in the adjustment effect (Figure 3). Most 95% HDIs for variable-level contributions are wide relative to their means. The only contribution showing a relatively consistent negative direction arises from the higher proportion of hormone receptor-positive patients at baseline in NeoSphere (71%) compared with HUS (58%) with mean contribution -0.03 with 95% HDI from -0.06 to 0.00 .

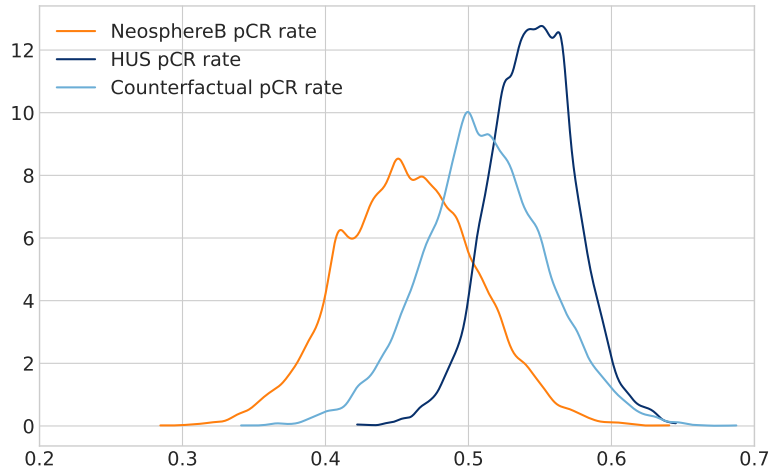


Figure 2: Estimated pCR rates for NeoSphere B arm, HUS and HUS with baseline adjustment.

1.2.2 KRISTINE comparison

The KRISTINE comparison is analyzed using the same framework as for NeoSphere. The posterior mean pCR rate in KRISTINE, $\hat{\theta}_{\text{RCT}}$, is 0.56 with a 95% HDI from 0.50 to 0.63. This does not indicate a clear difference from the HUS pCR rate $\hat{\theta}_{\text{RWD}}$; the posterior probability that $\hat{\theta}_{\text{RCT}} > \hat{\theta}_{\text{RWD}}$ is 0.62.

The baseline-adjustment shift estimated using Bayesian MRP with raking, $\Delta\hat{\theta}_{\text{adj}}$, has mean -0.12 with a 95% HDI from -0.23 to -0.01 (Figure 4). This corresponds to a counterfactual HUS pCR rate $\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}}$ with posterior mean

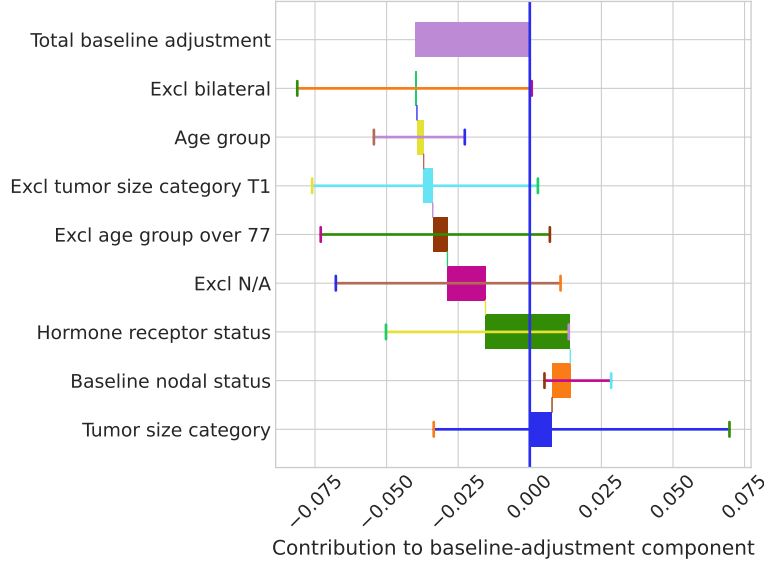


Figure 3: Shapley decomposition of the adjustment difference for the NeoSphere comparison. Bars represent mean Shapley contributions and error bars show 95% HDIs. Mean contributions sum to the total adjustment effect.

0.42 (95% HDI 0.33–0.51). The MAIC analysis yields a similar counterfactual estimate of 0.40 with a bootstrapped 95% confidence interval of 0.30–0.53, indicating broad agreement between the two adjustment approaches despite their different modeling assumptions. Posterior distributions for $\hat{\theta}_{\text{RCT}}$, $\hat{\theta}_{\text{RWD}}$, and $\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}}$ are shown in Figure 5.

Limited overlap between cohorts contributes to uncertainty in both adjustment approaches. The exclusion of nodal-positive patients reduces the pool of comparable cases, requiring extrapolation in the raking model and producing highly variable weights in MAIC. In the MAIC diagnostics, the mean effective sample size is approximately 30, and the 30 highest-weighted observations account for more than half of the total weight (Supplementary Figure 2).

The residual difference $\Delta \hat{\theta}_{\text{res}}$ (Figure 4) has mean 0.13 with a 95% HDI from 0.02 to 0.24. Despite substantial uncertainty, 94% of posterior mass supports a difference exceeding 5pp between $\hat{\theta}_{\text{RCT}}$ and $\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}}$. This suggests that factors not captured by baseline adjustment account for a sizable share of the observed gap.

The Shapley decomposition of $\Delta \hat{\theta}_{\text{adj}}$ indicates that the higher prevalence of hormone receptor-positive patients in KRISTINE (62%; Supplementary Table 3) relative to HUS (33%) corresponds to a mean -0.06 shift in the counterfactual pCR rate (95% HDI -0.09 to -0.03). This estimate likely understates the contribution of hormone receptor status, because patients excluded due to nodal-positive biopsy have a lower hormone receptor-positive proportion than

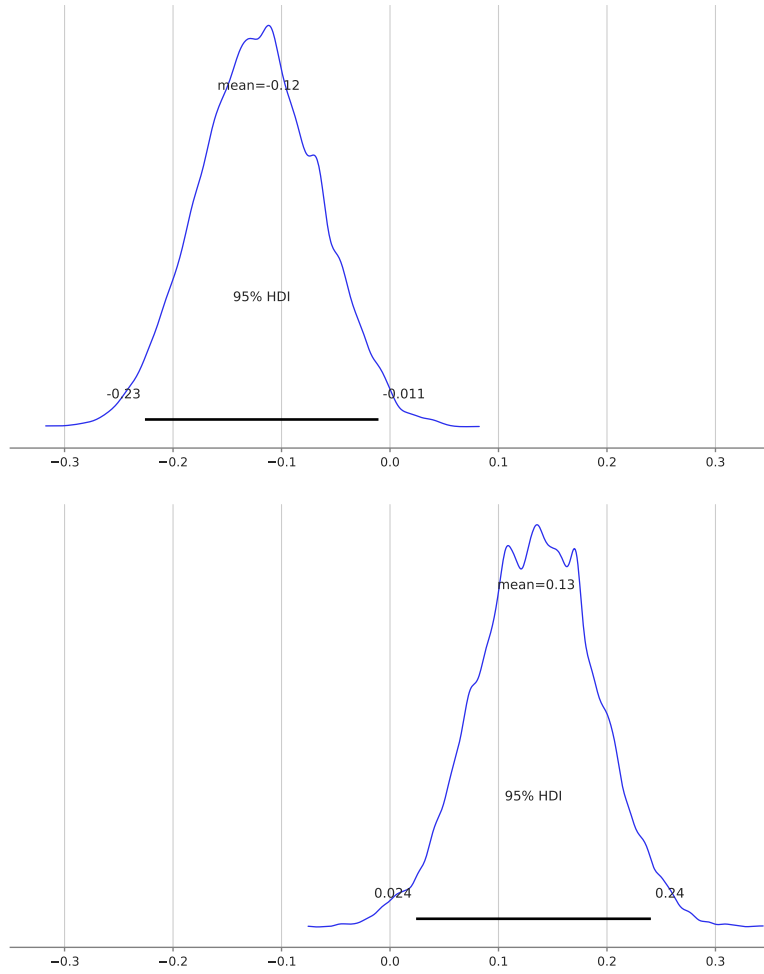


Figure 4: Top: Adjustment difference $\Delta\hat{\theta}_{\text{adj}}$ and Bottom: residual difference $\Delta\hat{\theta}_{\text{res}}$ for KRISTINE.

those retained (29% compared to 37%). As a result, part of the hormone-status effect may be absorbed by the nodal-biopsy exclusion variable. The Shapley value for nodal-biopsy exclusion has mean -0.04 (95% HDI -0.09 to 0.01), a direction that is difficult to interpret clinically and likely reflects this redistribution of attribution.

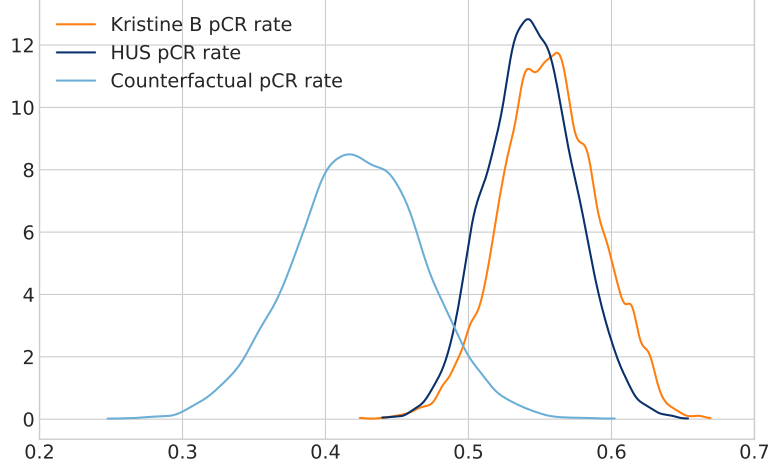


Figure 5: Estimated pCR rates for KRISTINE B arm, HUS and HUS with baseline adjustment.

1.2.3 HELEN-006 comparison

The unadjusted pCR rate $\hat{\theta}_{\text{RCT}}$ for the HELEN-006 arm B cohort is 0.57 with a 95% HDI from 0.52 to 0.63, comparable to the TCHP arm in the KRISTINE trial. The posterior probability that $\hat{\theta}_{\text{RCT}} > \hat{\theta}_{\text{RWD}}$ is 0.77, indicating moderate evidence for a higher pCR rate in HELEN-006 relative to HUS. However, the probability that this difference exceeds 5 percentage points is 0.31, suggesting limited support for a clinically large difference.

HELEN-006 provides the most detailed reporting of baseline characteristics (Supplementary Table 4), enabling adjustment for a broader set of variables than in the other comparisons. Posterior distributions for $\hat{\theta}_{\text{RCT}}$, $\hat{\theta}_{\text{RWD}}$, and $\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}}$ are shown in Figure 8, and the corresponding adjustment and residual differences are presented in Figure 7.

The adjustment difference $\Delta\hat{\theta}_{\text{adj}}$ has mean -0.07 with a 95% HDI from -0.17 to 0.03 . This corresponds to a counterfactual pCR rate $\hat{\theta}_{\text{RWD} \rightarrow \text{RCT}}$ with mean 0.48 (95% HDI 0.40–0.56). The MAIC estimate is similar at 0.49 (95% CI 0.40–0.57), providing convergent evidence across methods. The posterior probability that $\Delta\hat{\theta}_{\text{adj}}$ is positive is 0.10, indicating that a positive adjustment shift is unlikely. The residual difference $\Delta\hat{\theta}_{\text{res}}$ has mean 0.10 with a 95% HDI from 0.03 to 0.19.

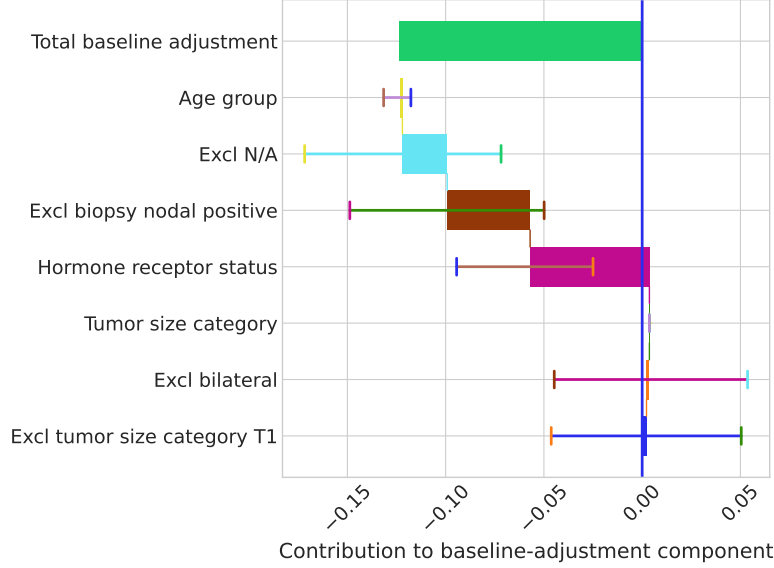


Figure 6: Shapley decomposition of the adjustment difference for the KRISTINE comparison. Bars represent mean Shapley contributions and error bars show 95% HDIs. Mean contributions sum to the total adjustment effect.

The Shapley decomposition (Figure 9) shows that the higher prevalence of hormone receptor-positive patients in HELEN-006 (57% vs. 33% in HUS) is the dominant contributor to the adjustment difference, with mean contribution -0.09 (95% HDI -0.13 to -0.05), accounting for roughly 40% of the total absolute Shapley contributions. Pre-treatment histology contributes positively (mean 0.04 , 95% HDI 0.02 – 0.06), largely reflecting the higher proportion of non-ductal tumors in HUS. Differences in HER2 IHC/ISH distribution also contribute modestly; the higher proportion of IHC2+/ISH+ cases in HELEN-006 (31% vs. 27% in HUS) corresponds to a Shapley contribution of -0.03 (95% HDI -0.05 to -0.01).

1.3 Sensitivity of results to NAT discontinuation

One potential contributor to the residual differences $\Delta\hat{\theta}_{\text{res}}$ in the KRISTINE and HELEN-006 comparisons is a higher proportion of patients in the HUS cohort who did not complete the planned NAT. To assess this, a binary treatment-discontinuation variable was added to the Bayesian MRP with raking model. The marginal distribution of this variable was aligned to the discontinuation proportions reported in the RCTs (NeoSphere B: 5/107 [3]; KRISTINE B: 18/221 [5]; HELEN-006 B: 23/337) and its influence on the estimates was evaluated.

Including treatment discontinuation in the adjustment set had minimal im-

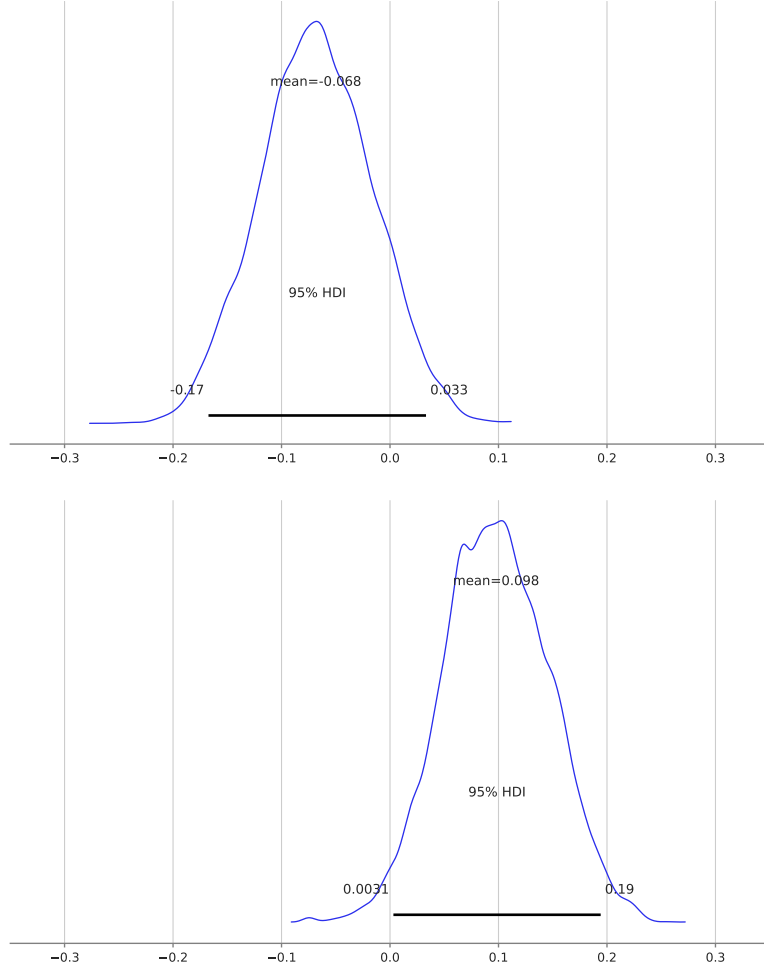


Figure 7: Top: Adjustment difference $\Delta\hat{\theta}_{\text{adj}}$ and Bottom: residual difference $\Delta\hat{\theta}_{\text{res}}$ for HELEN-006.

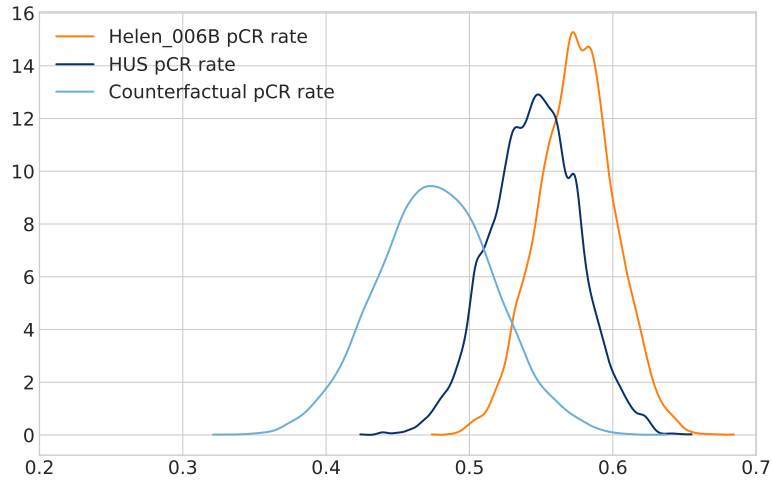


Figure 8: Estimated pCR rates for HELEN-006 B arm, HUS and HUS with baseline adjustment.

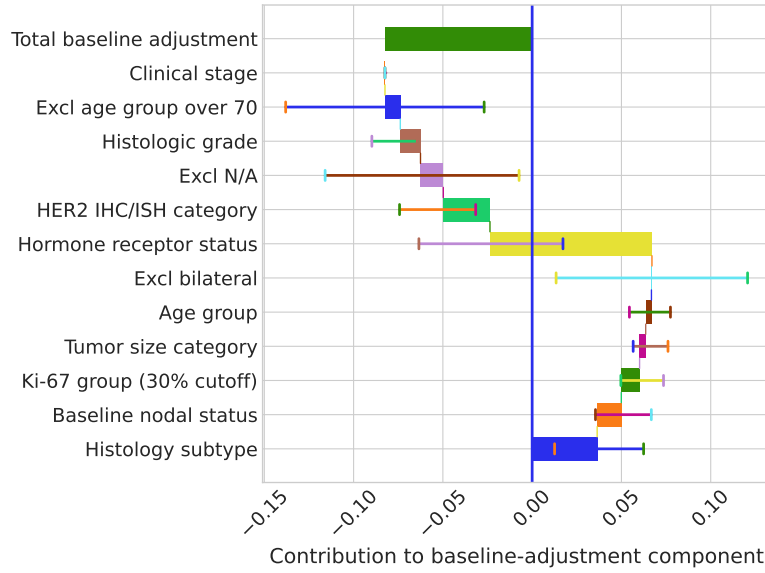


Figure 9: Shapley decomposition of the adjustment difference for the HELEN-006 comparison. Bars represent mean Shapley contributions and error bars show 95% HDIs. Mean contributions sum to the total adjustment effect.

Characteristic	N	Summary
Median age	337	50 (IQR 44-56)
T stage: T1 to T2	281	83%
T stage: T3 to T4	56	17
Nodal status: Positive	247	73%
Nodal status: Negative	90	27%
Bilateral: False	337	100%
Tumor grade: 1	1	< 1%
Tumor grade: 2	142	42%
Tumor grade: 3	194	58%
Histological type: Ductale	317	94%
Histological type: Lobulare	4	1%
Histological type: Other	16	5%
ER and/or PR positive: Positive	193	57%
ER and/or PR positive: Negative	144	43%
IHC: 2+ and ISH	104	31%
IHC: 3+	233	69%
Ki-67 index ≤ 30 %	75	22%
Ki-67 index > 30 %	262	78%
Disease stage: II	212	63%
Disease stage: III	125	37%

Table 2: Baseline characteristics for the HELEN-006 arm B (TCHP) population, restricted to variables used for adjustment and staging.

pact on the estimated residual differences. Across NeoSphere, KRISTINE, and HELEN-006, posterior means of $\Delta\hat{\theta}_{\text{res}}$ changed by less than 1 percentage point. Consistent with this, the outcome-model coefficients for treatment discontinuation in the KRISTINE and HELEN-006 analyses indicated only weak associations with pCR (Subsection 1.1). Although the discontinuation threshold differed in NeoSphere and was associated with a more negative coefficient, incorporating this variable did not materially alter the posterior distributions of $\Delta\hat{\theta}_{\text{adj}}$ or $\Delta\hat{\theta}_{\text{res}}$. These tests suggest that treatment completion is unlikely to be a major driver of the observed residual gaps

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